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Recommended Practice for Use of Langmuir Probes in Electric Propulsion Testing

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Presented in this work is a guide to standardize the methods of performing correct and accurate Langmuir probe measurements within the plasma environment produced by an electric propulsion device. The construction, installation, operation, and data interpretation of the full family of Langmuir probes are addressed including single Langmuir probes, double Langmuir probes, and triple Langmuir probes. Traditional time-averaged operation as well as high-speed operation are discussed as are the various pressure and external field limitations of Langmuir probes. This recommended practice is established from the vast collection of preexisting Langmuir probe theory and operational experience within the plasma environment of electric propulsion thrusters. Focus is placed on the application of Langmuir probes in the most common electric propulsion devices including Hall, ion, and pulsed thrusters. These electric propulsion devices all create flowing plasmas that require special probe construction and data analysis that is not needed in many other types of plasmas where Langmuir probes are commonly employed. The recommended practices are presented in a concise handbook style, reserving all theoretical derivations and details to the referenced works of their origination.

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Nomenclature

A_n = surface area of probe electrode n, where $\boldsymbol{A}$ denotes the vector normal to the surface, m^2
B = magnetic field strength, T
E = electric field, V/m
e = single electron charge, 1.60×10^{-19} C
I_p = probe current, A
$I_{e,n}$ = electron current to electrode n, A
$I_{e,\mathrm{sat}}$ = electron saturation current, A
$I_{i,\mathrm{sat}}$ = ion saturation current, A
$I_{+,n}$ = ion current to electrode n, A
Kn = λ/r_p, Knudsen number
k_B = Boltzmann constant, 1.38×10^{-23} J/K
L = length of cylindrical probe, where $\boldsymbol{L}$ denotes the vector collinear with the length, m
m_e = electron mass, 0.911×10^{-30} kg
m_i = ion mass, for Xenon mass is $131 \times 1.67 \times 10^{-27}$ kg
$n_{e,i,o}$ = electron, ion, and unperturbed plasma densities (quasi neutrality assumes all are equivalent), m^{-3}
q = species charge, C
R = ratio of double-probe electrode surface areas, A_2/A_1
$r_{L,e}$ = Larmor radius for electrons [usually $m_e\bar{v}_e/(|q|B)$], m
r_p = probe radius or characteristic dimension, m
$T_{e,i}$ = electron and ion temperature in electron volts (T_e is equivalent to $k_B T_{ek}$ [kelvin]/e), eV
V_B = probe bias with respect to ground, V
V_f = floating potential, with respect to ground V
V_n = potential of electrode n with respect to plasma potential, V
V_p = plasma potential with respect to ground, V
ΔV = voltage drop across probe current measurement shunt resistor, V
∂ = prefix to denote uncertainty of measured quantity
$\mathcal{E}$ = electron energy, eV
ϵ_o = permittivity of free space, 8.85×10^{-12} F/m
λ = mean free path [$1/(\sqrt{2}\sigma n)$ for Maxwellian distribution, $n\lambda_D^4$ for fully ionized plasma [3]], m
λ_D = electron Debye length, $\sqrt{\epsilon_o T_e/(n_e e)}$ m
σ = collision cross section, m^2
$\bar{v}_e$ = mean thermal electron speed [$\sqrt{8eT_e/(\pi m_e)}$ for Maxwellian distribution], m/s

I. Introduction

LANGMUIR probes are indispensable diagnostic tools that have been used to evaluate the plasma properties of electric thrusters since experiments with electric propulsion (EP) devices first began in the early 1960s [1]. The simplicity of Langmuir probe construction and operation (with straightforward methods of analysis) provides the experimentalist a powerful tool to measure the full state of the plasma that may be used to evaluate and identify performance losses, spacecraft interactions, erosion, stability of an electric thruster, and other plasmadynamic processes. Langmuir probes were first developed by Mott-Smith and Langmuir [2] in the 1920s as a method for measuring the temperature, density, and electric potentials of an ionized gas. Since then, the Langmuir probe has proved an important tool in the experimental study of plasmas of all types. The theory governing a Langmuir probe is well established [3–6] and has been expanded to cover Langmuir probe operation in high-pressure [7], strong magnetic fields, flowing, and unsteady plasma environments. The basic operation of a Langmuir probe involves biasing a small electrode immersed within a plasma and measuring the current drawn by the plasma for a range of bias voltages (see Fig. 1). The resultant current–voltage (I–V) characteristic is then analyzed by identifying various regions of ion current and electron current collection and using basic relations derived from energy conservation and Maxwell's equations (with limited assumptions including cold ions, highly mobile electrons, one-dimensional, Maxwellian, collisionless, and steady-state conditions) to compute the various plasma properties including electron temperature, electron and ion density, plasma potential, floating potential, and the electron energy

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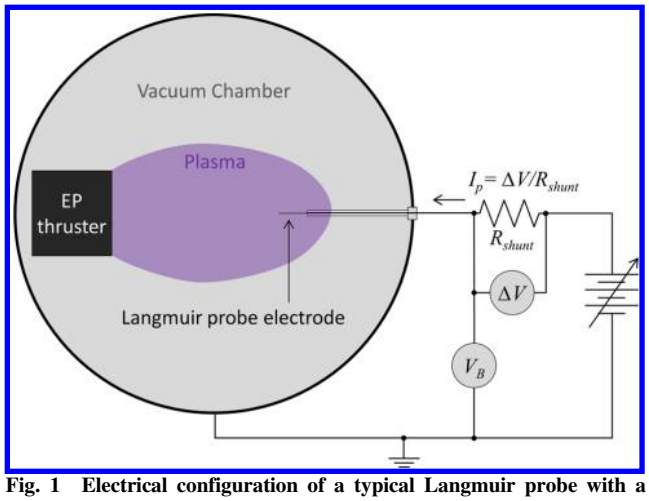

Fig. 1 Electrical configuration of a typical Langmuir probe with a cylindrical electrode biased to V_B drawing current I_p.

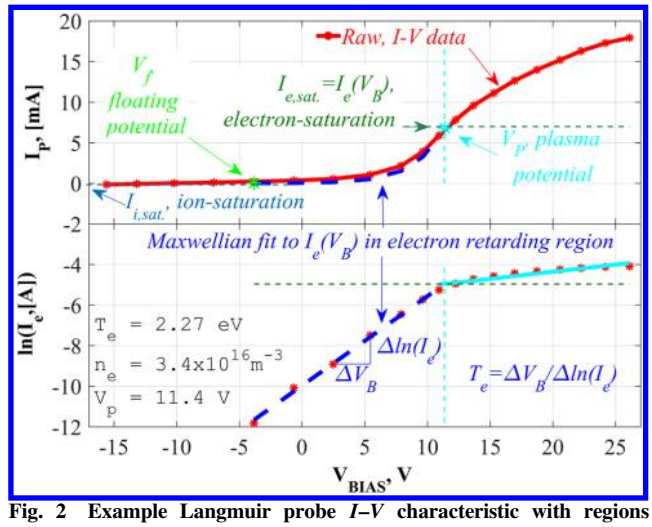

Fig. 2 Example Langmuir probe *I–V* characteristic with regions identified and analyzed. Data from the near-field plume of a Hall thruster, reprinted from [8], with the permission of AIP Publishing.

distribution function. All these properties are determined locally and simultaneously from a single *I–V* trace. With a small Langmuir probe (e.g., a thin wire less than 5 mm long) and a rapid bias sweep time ($\gtrsim \mu$s theoretically feasible [8]), a Langmuir probe offers excellent spatial and temporal resolution of six plasma parameters that is arguably unsurpassed by any other single plasma diagnostic. Yet the accuracy of Langmuir probe measurements is often quoted as being on the order of ±20–50%, due to a variety of limitations of the theory (and validity of assumptions) and measurement noise. However, this uncertainty can be reduced with the correct probe design, alignment, operation, and analysis, as is outlined in this document.

II. Single Langmuir Probe

A. Theory

Placing an electrically biased electrode into a plasma will generate a negative or positive current, depending on whether more ion or electron flux is attracted to the electrode. Scanning the electrode bias V_B over a wide range (e.g., −100 to +100 V for typical EP plume measurements) will result in the collected current varying from small negative values (microamperes typical in plume) at large negative voltages, when ions are predominantly collected, to larger positive values (greater than 10 mA typical in plume) at positive voltages, when mostly electrons are collected. The curve of electrode (or Langmuir probe) collected current at each bias voltage is referred to as a plasma *I–V* characteristic (Fig. 2), and analysis of this curve yields a wealth of information on the state of the plasma (density, temperature, potential, etc.).

The established theory of current collection to a Langmuir probe is actually quite simple if certain conditions are met. The most basic formulations of Langmuir probe theory require the following conditions to be met: 1) cold ions: $T_i/T_e \ll 1$, 2) electrons at or near thermal equilibrium (e.g., having a Maxwellian velocity distribution), 3) collisionless plasma: $Kn = \lambda/r_p \gg 1$, 4) electrostatic: $dB/dt \approx 0$, 5) nonmagnetized: $r_{L,e}/r_p \gg 1$, 6) quasi-neutral: $n_i \approx n_e$, and 7) isotropic and homogenous

Conditions 1–7 are not always observed in the plasma generated by EP devices, and in such cases, the theory of current collection can become significantly more complex. The method of Druyvesteyn (discussed in Sec. II.G) does not require condition 2 nor the assumption of a Maxwellian distribution and allows for the direct determination of the electron energy distribution function using derivatives of the *I–V* characteristic obtained by a single Langmuir probe. Also, the flowing plasma emitted by any EP thruster always violates condition 7, yet with proper probe alignment and operation (discussed throughout this document), this condition can indeed be realized. However, assuming that these criteria are satisfied, the theory of current collection by a Langmuir probe begins with consideration of the current resulting from the fluxes of electrons and ions passing through a planar surface A in space:

$$\text{plasma current(no probe; unperturbed)} = I = \frac{1}{4}eA(n_e\bar{v}_e - n_i\bar{v}_i) \quad (1)$$

where $\bar{v}_e$ and $\bar{v}_i$ are mean thermal electron and ion velocities. When a physical probe is immersed into the plasma, the natural flux of ions and electrons [as described by Eq. (1)] to its surface is usually perturbed as a plasma sheath is formed around the electrically biased probe with thickness on the order of the Debye length λ_D. The sheath is the name given to the spatial region where electron and ion densities vary from their unperturbed values as a result of the biased electrode. In the sheath, an electric field exists to establish a transition region between the applied probe potential and the unperturbed plasma potential V_p. Probes that are negatively biased with respect to the local plasma potential will repel electrons, whereas a positively biased probe will attract them (and vice versa for ions). When the bias is sufficient ($V_B \geq V_p$) to repel most ions and collect only electrons, the resulting current is referred to as the electron saturation current $I_{e,\text{sat}}$. Similarly, a probe bias sufficient to repel most electrons ($V_B \ll V_f$) and absorb only ions collects the ion saturation current $I_{i,\text{sat}}$ (where V_f is the bias at which the probe current is zero). The current collection between these regions is referred to as electron retarding and is portrayed in Fig. 2 as an exponential fit to the data between V_f and V_p. Analytically, one can use the equation of motion for a charged particle with Maxwell's equations as well as mass, momentum, and energy conservation in conjunction with the typical Langmuir probe assumptions listed previously (and assuming negligible electron mass), to arrive at the Boltzmann relation for a plasma [9]:

$$n = n_o \exp\left(\frac{\phi}{T_e}\right) \text{ or } n_e(V_B) = n_o \exp\left(\frac{V_B - V_p}{T_e}\right), \quad V_B \leq V_p \quad (2)$$

This relation implies that the plasma has a Maxwellian electron energy distribution, which is a common assumption for EP plasmas. Because ions are rarely in thermal equilibrium, a similar dependence for ion density with ion temperature is not usually observed. Using Eqs. (1) and (2) we then have the analytical electron current to an electrode of bias voltage V_B as

$$I_e(V_B) = I_{e,\text{sat}} \exp\left(\frac{V_B - V_p}{T_e}\right), \quad V_B \leq V_p, \quad I_{e,\text{sat}} = en_oA_p\sqrt{\frac{eT_e}{2\pi m_e}} \quad (3)$$

where n_o is the plasma density of the unperturbed plasma outside of the sheath surrounding the probe. The electron saturation current occurs when a probe is biased at the same potential as the plasma, $V_B = V_p$, and for this single bias, a sheath will not form about the probe. However, a relatively large electron current will be drawn out of the plasma ($I_{e,\text{sat}} = -\exp(+1/2)I_{i,\text{sat.}}\sqrt{(m_i/(2\pi m_e))} \approx -323 \cdot I_{i,\text{sat}}$

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for xenon, from Eq. (3) with thin-sheath ion current defined in the following Sec. II.B), and if this is a significant fraction of the total plasma discharge current, the probe may perturb the plasma [10].

Having now established the necessary relation for electron current collection by a biased electrode, the remainder of Langmuir probe analysis involves establishing an appropriate mathematical description for ion current collection, which is somewhat more complex than the electron component. The following sections will describe ion current collection models that are appropriate for various situations before summarizing a generalized approach for interpreting Langmuir probe I–V curves.

B. Thin-Sheath Ion Current Collection

When the thickness of the plasma sheath (thickness $\gtrsim \lambda_D$) surrounding a Langmuir probe electrode is negligible ($(r_p/\lambda_D) \gtrsim 50$), thin-sheath probe theory is applicable. Although the electron flux to a Langmuir probe is governed by the plasma thermal state [e.g., electron temperature in relation to the probe bias as described by Eq. (3)], the ion flux typically contains a negligible thermal component as described by assumption 1, from Sec. II.A. Indeed, the full derivation (see [4,5] or [6]) of ion current assumes stationary ions in regions unperturbed by the probe. Using the equation of motion for ions accelerating toward a negatively biased probe along with the Boltzmann relation [Eq. (2)] and the assumption of quasi neutrality necessitates the formation of a sheath about the probe electrode (within which ions outnumber electrons). The Poisson equation is then used to calculate the charge density distribution between the sheath edge to the probe surface and satisfying both energy and mass conservation invokes the Bohm condition wherein ions must move through the sheath edge at the Bohm speed [3] $v_B = \sqrt{eT_e/m_i}$. The total current to a Langmuir probe at a large negative bias (with thin-sheath assumptions) is then [9]

$$I_{i,\mathrm{sat}} = -\exp\left(-\frac{1}{2}\right) e n_o A_s \sqrt{\frac{eT_e}{m_i}}, \qquad \frac{r_p}{\lambda_D} \gtrsim 50 \tag{4}$$

The sheath area A_s is used for the formulation of the ion saturation current; however, with the thin-sheath criterion, one assumes $A_s \approx A_p$. This assumption of negligible ion sheath expansion is usually a poor one, and a slightly better result is attained using a Child–Langmuir sheath [3,11]:

$$A_s \approx \begin{cases} A_p \cdot \left(1 + \dfrac{x_s}{r_p}\right) \text{cylindrical probe} \\ A_p \cdot \left(1 + \dfrac{x_s}{r_p}\right)^2 \text{spherical probe} \end{cases},$$

$$x_s = \frac{\lambda_D \sqrt{2}}{3} \left[\frac{2(V_p - V_B)}{T_e}\right]^{3/4} \tag{5}$$

where r_p is the probe radius, and x_s is the sheath thickness. Equation (5) is only valid for an ion sheath devoid of all electrons ($V_p - V_B \gg T_e$), which is achieved when $V_B \ll V_f$. The Child–Langmuir sheath formulation also assumes that all ions present at the sheath edge are eventually collected by the probe. For a planar probe, sheath expansion augmented ion current collection is less pronounced, and in thin-sheath theory, $A_s \approx A_p$ is in most cases a good assumption. For thin sheaths about planar probes, sheath expansion can be envisioned as the sheath being displaced a small distance leading to a small increase in sheath area at the edges of the collector. Observations actually reveal an oblate ellipsoid shaped [13] sheath that transitions to hemispherical once the sheath size is comparable to the collector size (e.g., when $\lambda_D \gtrsim r_p$), as shown in Fig. 3.

C. Thick-Sheath Ion Current Collection

When the thickness of the plasma sheath surrounding a Langmuir probe electrode is large compared to the probe size ($(r_p/\lambda_D) \lesssim 3$), the collection of ions is referred to as orbital motion-limited (OML) because not all ions entering the sheath will be collected by the probe.

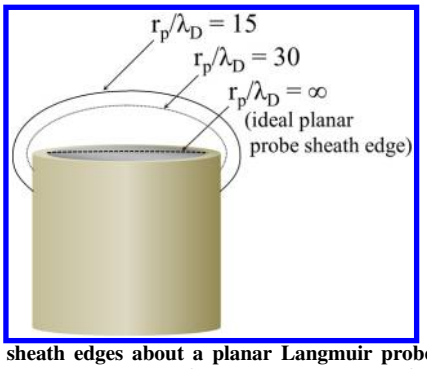

Fig. 3 Ion sheath edges about a planar Langmuir probe at bias of $V_B = V_p - 20 \cdot T_e$, for several r_p/λ_D ratios. Profiles for $r_p/\lambda_D = 15$ and 30 drawn to scale from PIC simulations in [12].

OML theory uses a similar approach as thin-sheath theory but includes angular momentum conservation and an ion energy distribution function to determine which ions reach the probe electrode. The resulting OML ion current no longer saturates to a fixed value [as in Eq. (4)] and is now a strong function of the probe bias but does not explicitly depend on the sheath thickness (because only ions within a certain impact parameter can reach the probe, a maximum ion current exists for a given probe bias and ion energy distribution) [2,4]:

$$I_i(V_B) \approx \begin{cases} \dfrac{e n_o A_p}{\pi} \sqrt{\dfrac{2e(V_p - V_B)}{m_i}} & \text{cylindrical probe} \\ e n_o A_p \sqrt{\dfrac{eT_e}{2\pi m_i}} \left(\dfrac{V_p - V_B}{T_e}\right) & \text{spherical and planar probes} \end{cases}$$

$$\frac{V_p - V_B}{T_e} \gg 1 \text{ and } \frac{r_p}{\lambda_D} \lesssim 3 \tag{6}$$

It is worth noting the lack of electron temperature dependence in the cylindrical probe OML ion current Eq. (6). This can simplify the analysis and reduce uncertainty in the determination of ion density but only if special care is taken in aligning the probe with the flow to ensure minimal wake asymmetries (see Sec. II.H).

D. Transitional Sheath Ion Current Collection

Often, the size of the probe is between the limiting cases of thin sheath and OML where a simple closed form for ion collection theory is not readily available. For such transitional ion sheath thicknesses (approximately $3 < (r_p/\lambda_D) < 50$), numerical solutions of the governing equations [14–16] or probe particle-in-cell (PIC) simulations [11,17] are used to parameterize the resulting ion current collection. The most complete transitional sheath results were computed by Laframboise [15], and these data have been since parameterized by several workers [18,19] into more compact and usable forms. The recent parameterization by Narasimhan and Steinbrüchel [19] defines the ion current, as given in Eq. (7) as a function of the probe bias and the Debye length from a power series fit to Laframboise's results (for $T_i/T_e = 0$):

$$I_i(V_B, \lambda_D) = e A_p n_o \sqrt{\frac{eT_e}{2\pi m_i}} a \left(\frac{V_p - V_B}{T_e}\right)^b$$

$$a = \begin{cases} 1.18 - 0.00080\left(\dfrac{r_p}{\lambda_D}\right)^{1.35} & \text{cylindrical} \\ 1.58 + \left[-0.056 + 0.816\left(\dfrac{r_p}{\lambda_D}\right)\right]^{-0.744} & \text{spherical} \end{cases}$$

$$b = \begin{cases} 0.0684 + \left[0.722 + 0.928\left(\dfrac{r_p}{\lambda_D}\right)\right]^{-0.729} & \text{cylindrical} \\ -0.933 + \left[0.0148 + 0.119\left(\dfrac{r_p}{\lambda_D}\right)\right]^{-0.125} & \text{spherical} \end{cases}$$

$$\frac{V_p - V_B}{T_e} > 1 \text{ and } 3 < \frac{r_p}{\lambda_D} < 50 \tag{7}$$

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This parameterization has been employed (and validated) in many plasma experiments and EP efforts [20–24] with excellent success. For planar probes, transitional ion sheath expansion has been examined through two-dimensional PIC simulations and a power series fit by Sheridan [12]:

$$
\begin{aligned}
I_i(V_B,\lambda_D) &= eA_p n_o\sqrt{\frac{eT_e}{2\pi m_i}}\,a\left(\frac{V_p-V_B}{T_e}\right)^b + \exp\left(-\frac{1}{2}\right)en_oA_p\sqrt{\frac{eT_e}{m_i}} && \text{planar}\\
a &= \exp\left(-\frac{1}{2}\right)\sqrt{2\pi}\left[2.28\left(\frac{r_p}{\lambda_D}\right)^{-0.749}\right] \approx 3.47\left(\frac{r_p}{\lambda_D}\right)^{-0.749} && \text{planar}\\
b &= 0.806\left(\frac{r_p}{\lambda_D}\right)^{-0.0692} && \text{planar}\\
3 &< \frac{V_p-V_B}{T_e} < 30 \quad \text{and} \quad 10 < \frac{r_p}{\lambda_D} < 45
\end{aligned}
\tag{8}
$$

The preceding planar probe model has been validated by a number of experiments [25], and an earlier analytic model [26] of an edge-modified Child–Langmuir sheath is also in good agreement [estimating Eq. (8) terms $a = (2\pi/3)(2)^{1/4}\sqrt{\exp(-1/2)}(r_p/\lambda_D)^{-1} \approx 3.20(r_p/\lambda_D)^{-1}$ and $b = 3/4$] with Sheridan's more detailed analysis. Sheridan's model assumes a free-floating planar probe and thus ignores wall sheath effects that may exist due to insulating material that might surround the planar probe edges. This edge-sheath distortion is often negligible at more negative probe biases and with larger probe radius to Debye length ratios. However, for planar probes in locations of relatively hot plasma (e.g., $T_e > 10$ eV), the probe bias needed to repel most high-energy tail electrons (and achieve the $[V_p - V_B]/T_e > 3$ requirement of Sheridan's power law fit) might become excessively large. For hotter plasmas (such as might be seen inside HET discharge channels [27]), wall secondary electron emission (SEE) becomes significant [28] and will cause the floating potential to move closer to the plasma potential. Once the SEE is fully saturated, the floating potential of the probe surface is only one or two electron temperatures below the plasma potential. This saturation effect is essential to the function of emissive probes, which employ a thoriated tungsten filament that, when resistively heated to full SEE saturation, floats to within 1–2 T_e factors from the local plasma potential as in Eq. (9) [29]:

$$V_{f,\text{hot-emissive-probe}} \approx V_p - T_e \tag{9}$$

E. Electron Current Collection Above Plasma Potential

One of the advantages of using the electron saturation current to determine electron density [Eq. (3)] is the independence of the sheath size (as a probe biased to V_p is sheathless); however, biasing a probe above the plasma potential will exhibit sheath expansion (theoretically) as shown in Fig. 4. The "knee" of the I–V trace (location of the plasma potential) is easily identified in the lower plot of Fig. 4 as the region where the exponential growth (linear in log space) flattens out.

In the thick-sheath (OML) limit, the electron current increases as the square root of the bias for cylindrical probes and grows linearly with bias for spherical (and nonguarded planar) probes. Thin sheaths ($(r_p/\lambda_D) \gtrsim 50$) also still expand when $V_B > V_p$, but to a lesser degree than OML, because any expanded extra sheath area is significantly smaller than the probe area (however, this expansion follows stronger power laws, namely 3/4 and 3/2 powers of probe bias above the plasma potential for cylindrical and spherical probes, respectively). This behavior lends itself as a visual aid to locating the plasma potential from the raw I–V characteristic as the transition from exponential current growth to linear or 1/2 power growth results in a visible knee or inflection point (however, obscuration of this knee can occur as discussed in Secs. II.H.2, II.I). The included ideal "guarded" planar probe trace in Fig. 4 shows no increase electron current when $V_B > V_p$, which is achieved with a configuration as in Fig. 5.

For guarded or unguarded planar probes directed at an electric propulsion thruster (this is not the recommended orientation for the measurement of electron or ion density, electron temperature, or plasma potential as it violates the assumption that ions enter a sheath at the Bohm speed), the primary ion beam generated by the thruster can be reflected at a sufficiently high bias.

Figure 6 qualitatively illustrates this ion beam reflection effect, which has been observed in low-energy ion beams [30,31] as well as with higher-energy ion beams in Hall thruster studies [27]. The probe electrode alignment in Fig. 6 is included here because it may be unavoidable in certain scenarios (e.g., wall-mounted probes [27]), and it does provide a method of estimating the incident ion beam energy. Ion energy distribution functions should be obtained with

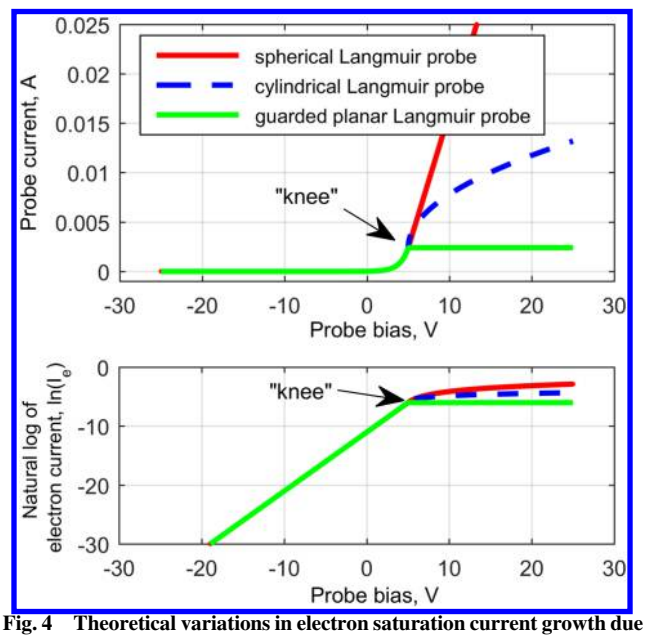

Fig. 4 Theoretical variations in electron saturation current growth due to sheath expansion for Langmuir probes in a nonflowing plasma (thick-sheath OML).

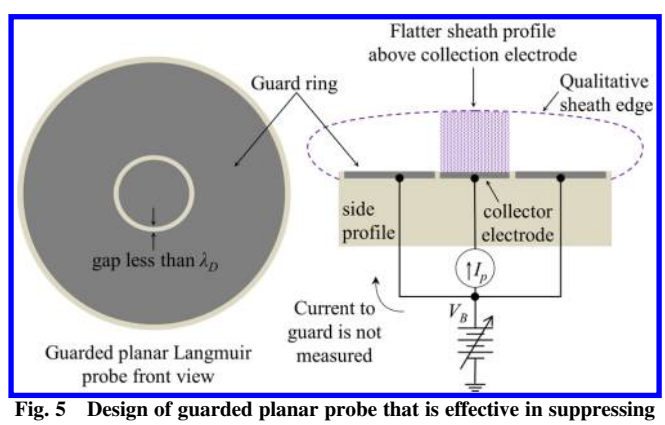

Fig. 5 Design of guarded planar probe that is effective in suppressing sheath expansion for interelectrode gaps $< \lambda_D <$ over all probe dimension.

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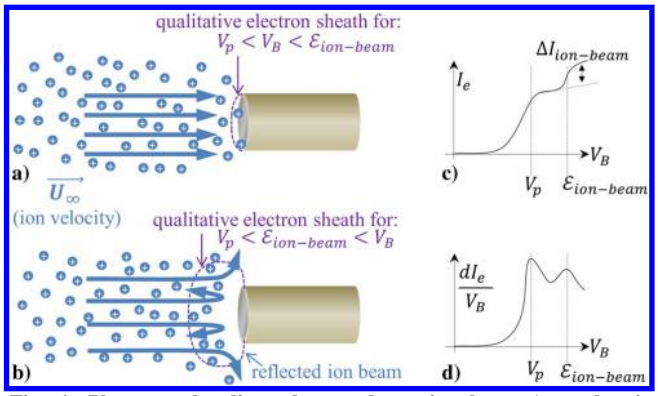

Fig. 6 Planar probe directed towards an ion beam (e.g., electric thruster) biased to reflect the ions (Fig. 6b) creates an ion stagnation zone with dramatically increased sheath area and electron current (Fig. 6c).

either a retarding potential analyzer (RPA), electrostatic analyzer, or $\mathrm{E}\times\mathrm{B}$ probe. The presence of an ion beam is most apparent when plotting the first derivative of the electron current versus probe bias as a second peak in the profile centered on the most-probable beam energy (shown qualitatively in Fig. 6d). However, operation of a Langmuir probe at biases well above the plasma potential will draw significant electron current from the plasma, which could alter the local plasma potential (due to insufficient balancing flux of ions), give rise to excessive probe heating, or induce an electrostatic ion wave instability [32].

When a transitional electron sheath exists ($5 < r_p/\lambda_D < 100$), and the probe is properly aligned with the plasma flow (see Sec. II.H), fits to Laframboise's numerical analysis [15] for cylindrical electron saturation current collection may be applied. This approach is similar to the transitional ion sheaths discussed in Sec. II.D with the electron saturation current as [33]

$$I_{e,\mathrm{sat}}(V_B,\lambda_D,T_i/T_e) = eA_p n_o\sqrt{\frac{eT_e}{2\pi m_e}}\left(a_e+\frac{V_B-V_p}{T_e}\right)^{b_e}$$

$$a_e = -2.8+\left(\frac{T_i}{T_e}\right)\left\{5.1+0.135\left[\ln\left(\frac{r_p}{\lambda_D}\right)\right]^3 - \frac{2.8}{\ln(r_p/\lambda_D)}\right\}+\frac{2.8}{\ln(r_p/\lambda_D)}$$

$$b_e = \frac{2.9}{\ln(r_p/\lambda_D)+2.3}+0.11\left(\frac{T_i}{T_e}\right)^{0.65}-0.38$$

$$\frac{V_B-V_p}{T_e} > 3 \quad \text{and} \quad 5<\frac{r_p}{\lambda_D}<100 \tag{10}$$

These fits include an ion-to-electron temperature ratio dependence T_i/T_e) that can be neglected with most EP plasmas. It should be reemphasized that this formulation is not needed to determine the electron density if the plasma potential is known [because Eq. (3) provides n_e from the probe I–V characteristic at V_p for any r_p/λ_D]. However, if the plasma potential is unknown (e.g., there does not exist a well-defined peak in dI_e/dV_B nor knee in the I–V characteristic), then Eq. (10) might prove useful in determining both n_e and V_p based on the growth of the electron current at higher biases [likely requiring an iterative approach because both n_e and V_p are unknowns in Eq. (10)].

F. Algorithm for Langmuir Probe Analysis (Single Probe)

The following procedure will yield the plasma properties obtainable with a single Langmuir probe, which follows the typical assumptions as listed in Sec. II.A.

1) Scan the Langmuir probe voltage (see Fig. 1) from a large negative voltage to a large positive voltage while measuring both the probe bias voltage V_B and the probe current I_{probe}. The scan should span through both ion saturation (where the current is a small slowly varying negative current) and just into electron saturation, which occurs at the plasma potential V_p. A sweep from −50 to +20 V is usually sufficient for the far-field plume of a typical Hall thruster. See [34] for example Hall thruster plume plasma potential maps in the far field and [35] for near-field and internal maps.

2) Locate the voltage where the probe current is closest to 0 A, recording that bias as the floating potential V_f.

3) Fit a line to the I–V characteristic for all voltages below V_f and define this as the preliminary ion saturation curve, $I_{i,\mathrm{sat.}}(V_B) = m_{i,\mathrm{sat}}V_B + b_{i,\mathrm{sat}}$, where $m_{i,\mathrm{sat}}$ and $b_{i,\mathrm{sat}}$ are the least-squares determined slope and y intercept. This linear fit approximates the slight increase in ion current observed at negative probe biases (with respect to plasma potential). A slightly more accurate fit could incorporate a Child–Langmuir sheath (electron free) by fitting to the 3/4 and 3/2 powers of probe bias for cylindrical and spherical probes, respectively.

4) Compute the electron current as $I_e(V_B) = I_{\mathrm{probe}}(V_B) - I_{i,\mathrm{sat}}(V_B)$.

5) Compute the first derivative of the electron current, $dI_{e,\mathrm{smoothed}}/dV_B$, applying numerical smoothing if needed to remove measurement noise. Locate the maximum value of this derivative and label that bias as the plasma potential V_p. If this method fails (namely, if no maximum exists), then the formula in the alternate step 6 may be used to estimate V_p once V_f and T_e are known.

6) Plot the natural logarithm of the electron current versus probe bias ($\ln I_e$ versus V_B) and compute the least-squares determined slope of the linear region between V_f and V_p. The electron temperature is the inverse of this slope: $T_e|_{\mathrm{slope-method}} = (d\ln I_e/dV_B)^{-1}$. Or, alternately, if the electron temperature cannot be determined from the preferred (more accurate) slope method, then the floating and plasma potentials (from a cold probe without SEE) can instead be used to estimate the electron temperature:

$$T_e|_{\mathrm{potential-method}} = (V_p-V_f)/\ln\left(\sqrt{\frac{m_i}{2\pi m_e}}\right) \approx (V_p-V_f)/5.28 \ \text{(for xenon ions)}$$

7) Compute the electron density as $n_e = (I_{e,\mathrm{sat}}/eA_p)\sqrt{(2\pi m_e)/(eT_e)}$ where $I_{e,\mathrm{sat}} = I_e(V_B = V_p)$.

8) Compute the preliminary ion density as $n_i = -(\exp(1/2)I_{i,\mathrm{sat}}/eA_s)\sqrt{(m_i)/(eT_e)}$.

a) Use thin-sheath assumption: $A_s \approx A_p$.

b) Or use Child–Langmuir sheath [Eq. (5)].

9) Compute the Debye length as $\lambda_D = \sqrt{\epsilon_o T_e/(n_o e)}$ using either the electron or ion density for n_o [using n_e from Eq. (7) if available because it is unaffected by sheath size uncertainty].

10) Use the probe radius to Debye length ratio r_p/λ_D) to determine applicable sheath-expansion-corrected ion density equation.

a) Thin sheath when $r_p/\lambda_D \gtrsim 50$; ion density computed in step 8a is correct; negligible sheath expansion exists.

b) Transitional sheath when $3 < r_p/\lambda_D < 50$; ion density needs correction for sheath expansion according to terms a and b from Eq. (7) or Eq. (8):

$$n_i = \frac{1}{aA_p}\sqrt{2\pi m_i}e^{-(3/2)}T_e^{b-(1/2)}\left[-\frac{dI_{\mathrm{probe}}^{1/b}}{dV_B}\right]^b \tag{11}$$

To compute ion density from Eq. (11), plot $I_{\mathrm{probe}}^{1/b}$ versus V_B and use a method of least-squares to linearly fit a line (or just take the average slope) from most negative probe bias to the plasma floating potential. Insert that slope as the term in brackets $\left[-dI_{\mathrm{probe}}^{1/b}/dV_B\right]$ along with the other known terms to compute the sheath corrected ion density.

c) OML sheath when $3 \gtrsim r_p/\lambda_D$; ion density needs correction for sheath expansion according to Eq. (6). Equation (11), for ion density, works for cylindrical probes with $a = (2\sqrt{\pi})$, $b = 1/2$ and for spherical and planar probes with $a = b = 1$.

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11) If a sheath correction was applied (i.e., nonthin sheath), then iterate steps 4–10 using the updated $I_{i,\mathrm{sat}}(V_B) = I_i(V_B)$ [from Eqs. (6–8)] until convergence is achieved (usually only a few loops are required). Iteration is needed to refine the correction because sheath expansion is a function of the plasma density (used to compute λ_D), electron temperature, and plasma potential.

The preceding steps are easily automated in any programming language or plotting software. The computation of plasma potential (using maximum of dI_e/dV_B) in step 5 is generally successful and easy to automate in code. Another common method is to look for the knee in the $\ln(I_e)$ versus V_B plot (shown in lower traces of Figs. 2 and 4), which is the point that shows a sharp change in the slope that occurs as the exponential current growth enters electron saturation current. The probe current at biases greater than the plasma potential will still continue to grow because an electron sheath will form and expand as the bias is increased (see Sec. II.E). If possible, increased accuracy for the measurement of electron density and electron temperature may be obtained from moments of the electron energy distribution function (EEDF) as discussed in the following section.

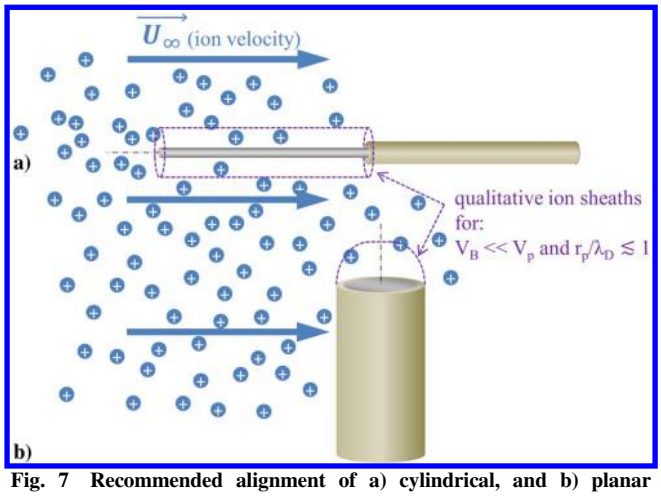

Fig. 7 Recommended alignment of a) cylindrical, and b) planar Langmuir probes with respect to supersonic plasma flow velocity vector.

G. Measuring the Electron Energy Distribution Function

The Langmuir probe approaches described in preceding sections assumed a Maxwellian electron energy distribution; indeed most EP plasmas are analyzed assuming such a distribution. A Maxwellian distribution originates from sufficient collisions having occurred to equilibrate electron thermal motions, yet the bulk of an EP plume is considered nearly collisionless. Thus, the assumption of a Maxwellian distribution implicitly assumes the existence of a collisional region within a given EP device, such as the high-density ionization region (e.g., the Hall current region of Hall thrusters). Assuming that this ionization region thermalizes the electrons, they are then assumed to expand outward [according to an Eq. (2) type Boltzmann relation] retaining their original Maxwellian distribution. However, in cases of magnetized electrons, large anisotropic ion velocities, plasma wall effects, and anomalous electron mobility, plasma electrons can deviate from a Maxwellian distribution [36–39]. A Langmuir probe can be used to measure the EEDF and thereby assess its deviation from Maxwellian. The Druyvesteyn method [40] provides the EEDF (in units of per (cubic-meter times electron volt)) from the second derivative of the electron current [11]:

$$f(\mathcal{E} = V_p - V_B) = \frac{2}{e^2 A_p}(2m_e e\mathcal{E})^{1/2}\frac{d^2 I_e}{d\mathcal{E}^2} \tag{12}$$

Once the EEDF is obtained, the electron density and effective electron temperature may be computed from the following moments of the distribution function:

$$n_e = \int_0^\infty f(\mathcal{E})\,\mathrm{d}\mathcal{E} \quad T_{e,\mathrm{eff.}} = \frac{2}{3n_e}\int_0^\infty \mathcal{E} f(\mathcal{E})\,\mathrm{d}\mathcal{E} \tag{13}$$

Because the EEDF is computed in the electron retarding region (e.g., at biases below the plasma potential), there is no electron sheath, and the use of electron current requires the removal of ion current (and hence removal of ion sheath effects), thereby eliminating dependence on the probe size to Debye length ratio in Eqs. (12) and (13). In addition, the original derivation by Druyvesteyn showed invariance with probe geometry, thus validating Eqs. (12) and (13) for spherical, cylindrical, planar, and all other convex surface electrodes. Unfortunately, one of the Druyvesteyn method assumptions is an isotropic plasma, which an EP plasma often violates. Beamlike anisotropy can give rise to a drifting distribution, one that still retains $f(\mathcal{E}) \propto d^2 I_e/d\mathcal{E}^2$ [not the exact form of Eqs. (12) and (13)] but contains additional anisotropy coefficients [39]. However, the anisotropic effect on electron current collection (and the EEDF) is mitigated when the anisotropic drifting beam component is parallel to the electrode surface. This arrangement should be achievable with a thin-sheath planar or cylindrical Langmuir probe oriented with the normal of its surface (ignoring end effects for a cylindrical probe) perpendicular to the electron drift direction (see Fig. 7). The resulting three-dimensional (3-D) EEDF only measures the nondrifting EEDF component perpendicular to the probe surface. One severe disadvantage of the Druyvesteyn method is the use of the second derivative of the probe I–V data. Plasmas in general are not quiescent, and plasma measurements are often noisy, requiring significant averaging to increase the signal-to-noise ratio to acceptable levels. Taking second derivatives from such data is often futile unless extreme care is taken in the measurement to average enough data and eliminate all possible measurement or instrument bias errors including the following.

1) Improper or unaccounted probe-plasma circuit impedances (see [41] for more details).

2) Capacitive displacement current skew (due to finite dV_B/dt during probe I–V sweeps).

3) Improper averaging or fitting techniques; for example, averaging windows that are too large will remove sharp EEDF features and flatten or cause artificial "Druyvesteynization" of the distribution near the plasma potential).

4) Unsteady or drifting plasma properties (e.g., distortion of I–V due to RF discharges or a drifting that varies while I–V data are collected).

5) Incorrectly subtracted ion current. Many authors [42] argue that, because $|I_e| \gg |I_i|$ (near the plasma potential), one may use $I_p = I_e + I_i \approx I_e$ for computing the second derivative. This approximation appears valid for electron energies below $7.5T_e$ [41] but breaks down at higher energies when $(d^2 I_i/d\mathcal{E}^2) \gg (d^2 I_e/d\mathcal{E}^2)$.

6) Lack of sufficient circuitry and data conversion bandwidth for obtaining accurate measurements. Notably, using large resistors (as shunts or voltage dividers) can prove problematic because even small stray capacitances can lead to passive (resistor-capacitor) low-pass filter formation with cutoff frequencies as low as 10 Hz (thereby causing data collected at faster than 10 samples per second to be in significant error).

7) Lack of a sufficiently small, nonperturbing probe and probe holder.

These errors can prove challenging to fully mitigate because they can involve measurement subtleties rarely appreciated. For example, with regard to error source 1, the full Langmuir probe-plasma measurement circuit includes amplifier output impedance (which should be few small, ideally 0 Ω for dc measurements), cable resistance (and reactance if high-speed measurements are being made), current measurement shunt resistance, probe surface contact resistance, chamber/reference electrode surface, and sheath resistance. This later resistance is often overlooked because single Langmuir probes for EP are usually tested in metallic vacuum chambers, which provide abundant surface area to reduce the effective wall contact and sheath resistance (which acts as the reference electrode to return the probe current path). However, testing in very large vacuum chambers or those with nonelectrically conductive chamber walls can increase the chamber/reference

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electrode resistance significantly (causing a voltage drop at the probe tip that is not easily measured). In such cases, an explicit reference electrode (grounded or floating in an asymmetric double-probe configuration) should be included near the main Langmuir probe (and the reference probe should have area larger than active Langmuir probe by at least a factor of $\sqrt{m_i/m_e}$).

In a manner analogous to retarding potential analyzers (RPAs) [43], which measure ion energy distribution functions, properly aligned planar and cylindrical Langmuir probes can be used to determine the one-dimensional (1-D) EEDF for a plasma from the first derivative of the retarded electron current. A 1-D EEDF only measures the electron energy distribution in the coordinate that is normal to the probe surface (e.g., perpendicular to a planar collector or radial for a cylindrical collector). These 1-D EEDFs will not give any information on electrons whose trajectories are parallel with the collector surface. For a planar Langmuir probe (aligned as shown in Fig. 7b), the 1-D EEDF is simply as follows [39,44,45].

Planar probe 1-D EEDF:

$$f_{1D}(\mathcal{E} = V_p - V_B) = \frac{1}{eA_p}\left(\frac{2m_e}{e\mathcal{E}}\right)^{1/2}\frac{dI_e}{d\mathcal{E}}$$
$$f_{1D}(\mathcal{E}_{k-(1/2)} = V_p - V_{B,k-(1/2)}) = \frac{1}{eA_p}\left(\frac{2m_e}{e\mathcal{E}_{k-(1/2)}}\right)^{1/2} \times \frac{(I_{e,k} - I_{e,k-1})}{(\mathcal{E}_k - \mathcal{E}_{k-1})} \qquad \text{(discretized)} \tag{14}$$

The inverse square root $\mathcal{E}$ term is prevented from becoming singular by use of the included discrete derivative, which uses the midpoint between the first two $\mathcal{E}$ points. The distribution function $f_{1D}(\mathcal{E})$ in Eq. (14) has been normalized suitably to enable the calculation of electron density and effective electron temperature from the distribution function in the same manner as the Druyvesteyn method [using Eq. (13)]:

$$n_e = \int_0^\infty f_{1D}(\mathcal{E})\,\mathrm{d}\mathcal{E} \quad T_{e,\text{eff.}} = \frac{2}{n_e}\int_0^\infty \mathcal{E} f_{1D}(\mathcal{E})\,\mathrm{d}\mathcal{E} \tag{15}$$

This first derivative EEDF method for planar probes has only limited validation in EP [46] and other plasmas [43,47], but the general theory is well validated in the considerable RPA data that abound for EP devices.

For cylindrical probes, a similar probe alignment symmetry (as in Fig. 7a) allows for a determination of the radially directed 1-D EEDF. A closed-form explicit distribution [such as Eq. (14)], other than obtained via the Druyvesteyn approach, as with Eq. (12), is not available. However, an implicitly propagated discretized approximation is available [45] using Eq. (16). In this discretized solution, the 1-D EEDF is computed marching backward from the most negative probe biases (e.g., taking $k = 0$ for $\mathcal{E} = \text{maximum measured} = V_p - V_{B,\text{min.meas.}}$) using previous distribution points to backward predict subsequent distribution points.

Cylindrical probe (discretized) 1-D EEDF:

$$f_{1D}(\mathcal{E}_{k+(1/2)} = V_p - V_{B,k+1/2}) = \begin{cases} 0 & k = 0 \\ \frac{3\pi I_e(\mathcal{E}_1)}{2A_p(\mathcal{E}_0-\mathcal{E}_1)^{3/2}}\sqrt{\frac{m_e}{2e^3}} & k = 1 \\ \left\{\frac{3\pi I_e(\mathcal{E}_k)}{2A_p}\sqrt{\frac{m_e}{2e^3}} - \sum_{j=1}^{k-1} f_{1D}(\mathcal{E}_{j-1/2})\left[(\mathcal{E}_j - \mathcal{E}_k)^{3/2} - (\mathcal{E}_{j-1} - \mathcal{E}_k)^{3/2}\right]\right\}(\mathcal{E}_{k-1} - \mathcal{E}_k)^{-3/2} & k > 1 \end{cases} \tag{16}$$

This approach for cylindrical probes should be less noise prone than the Druyvesteyn method, but the implicit form of the distribution can experience numerical instability. The formulation of the cylindrical probe 1-D EEDF in Eq. (16) is proportional to $\propto |\mathcal{E}_k - \mathcal{E}_{k-1}|^{-3/2}$, whereas the Druyvestyn method (full 3-D) EEDF in Eq. (12) is $\propto |\mathcal{E}_k - \mathcal{E}_{k-1}|^{-2}$; thus, the bias measurement noise is still amplified to a similar degree with both methods. The numerator for Eq. (16) is approximately $\propto |I_k - I_{k-1}|$, whereas the Druyvestyn method is $\propto |I_k - I_{k-1}|^2$, thus gaining slightly better current measurement noise immunity with the 1-D EEDF approach.

The correct cylindrical probe alignment (Fig. 7a) ensures an isotropic 1-D distribution in the radial direction (which [45] requires), and if this symmetry is broken, then Eq. (16) is invalid. Also, the discretized forms of the electron energy distribution functions in Eqs. (14) and (16) require equally spaced probe bias points (one may need to fit a spline to the data to attain a set of equally spaced bias points). Density and effective electron temperature may then be computed from Eq. (16) using the same method for 1-D EEDFs used with planar probes. Although the cylindrical 1-D EEDF is included here for completeness, insufficient data exist at this time to fully validate this method; thus, it is recommended that, if cylindrical probes are used, then the Druyvesteyn method (with proper alignment as in Fig. 7a) should be used to compute the EEDF.

Overall, if EEDFs are to be measured in an EP environment, it is recommended that properly aligned planar probes are employed because the theory [from either Eq. (12) or Eq. (14)] of computing EEDFs from planar I–V data is still valid (and has been experimentally validated) in the anisotropic environment of an EP thruster.

H. Flowing Plasma Considerations

Generally, EP plasmas are supersonic, with directed plasma flow well in excess of the Bohm speed:

$$\text{Supersonic plasma Mach number: } \frac{U_\infty}{U_{\text{Bohm}}} > 1, \quad \text{where } U_{\text{Bohm}} \approx \sqrt{eT_e/m_i} \tag{17}$$

Alignment of the probe collection surface parallel to this velocity of ion ram flux largely mitigates most flowing plasma effects. Heavy supersonic ions are not easily deflected by a Langmuir probe, and thus any ion-beam incident collection surface will receive a roughly constant ram flux of ions proportional to the ion velocity. Extremely large probe biases could deflect the ions (see Sec. II.E), but the resulting electron current might melt the probe or lead to sheath breakdown. The recommended probe arrangement is shown in Fig. 7 for cylindrical and planar probes. The cylindrical probe requirement (Fig. 7a) that the velocity vector is collinear with the probe axis may be difficult to maintain without a priori knowledge of the flow streamlines. The planar probe requirement (Fig. 7b) is easier to guarantee for a planar probe lying on any plane intersecting the axis of symmetry (because most EP plasmas are axisymmetric). A typical Hall thruster with a 300 V discharge accelerates singly ionized ions to roughly $U_\infty \approx \sqrt{2 \cdot e \cdot V_D/m_i} = 21$ km/s, whereas the Bohm speed for a 1.5 eV region of the plume is just 1050 m/s. Thus, the plasma Mach number is often much larger than unity for typical EP plasmas, whereas for the vast majority of plasma research (e.g., Tokomaks, plasma processing, etc.), the ion flow is usually subsonic or close to sonic. This crucial difference with EP plasma diagnostics is not always appreciated, and many EP papers are published that neglect flowing plasma effects. This omission most likely arises from the application of the standard Langmuir probe theory shown in many textbooks, theories that often fail to mention flowing plasma effects because they are rarely encountered outside of EP research. The simple step of selecting the correct probe alignment (Fig. 7) is all that

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is needed to correctly operate a Langmuir probe in a flowing plasma using the same theory as a nonflowing plasma.

1. Electron Temperature and Electron Saturation with Flow

The electron temperature T_e is independent of both probe angle [48] (or shape) and flow velocity [49], provided that the flow velocity is less than the electron thermal velocity $U_\infty \ll \sqrt{eT_e/m_e}$. Electron thermal velocities are usually well in excess of EP flow velocities; for example, a 1.5 eV plasma has an electron thermal velocity over 510 km/s. However, improper probe alignment can result in a failure to achieve traditional electron current saturation. This effect can transition the exponential region of electron current growth to a linear region [50] that can yield falsely high electron temperatures if the $\ln I_e$ versus V_B inverse slope method includes voltage points near the plasma potential. It is recommended that points a few T_e below V_p are used:

$$\text{Slope} - \text{method distortion prevention:}\quad T_e|_{\text{slope-method}} = \left(\frac{d\ln I_e}{dV_B}\right)^{-1} \qquad V_f \lesssim V_B \lesssim V_p - 2\cdot T_e \tag{18}$$

One reason for electron saturation becoming less distinct (i.e., electron current continues to increase) in EP plasmas is that the large velocities of primary beam ions (greater than 10 km/s or greater than 100 eV/Xe$^+$ is typical) directed toward a probe are largely unaffected by positive probe biases [50,51] and prevent the typical Debye shielding seen in stationary plasmas. This lack of an ion-deficient sheath exists only with probes oriented with collection surfaces normal with the flow vector because beam ions can only be pushed away (to allow electron sheath formation) if the probe is biased at an excessively large voltage (e.g., a bias of 100 V or more above the plasma potential is required to push away 100 eV ions; see Sec. II.E). Local oscillations (common in many types of EP devices) of the plasma properties can also result in less distinct electron saturation [4].

2. Ion Saturation with Flow

For cylindrical probes (of length L) aligned with the flow direction (as in Fig. 7), end-effect deviations in ion saturation are minimal when the probe length to Debye length ratio is much greater than the plasma Mach number [52]:

$$\text{Negligible cylindrical probe end effect:}\quad \frac{L}{\lambda_D} \gg \frac{U_\infty}{U_{\text{Bohm}}}, \qquad \boldsymbol{L} \| \boldsymbol{U}_\infty \tag{19}$$

For thin-sheath probe theory (which assumes $L \gg r_p \gg \lambda_D$), end effects can usually be neglected in computing ion density for flow-aligned Langmuir probes. In highly supersonic plasma flows, negatively biased probes oriented toward the flow have been shown [53] to predominantly collect a ram ion current that is merely equivalent to the flux of ions swept by the probe area intersecting the flow vector [2]:

$$\text{Ram ion current:}\ I_{\text{ion-ram}} = en_i \boldsymbol{A}\cdot\boldsymbol{U}_\infty \tag{20}$$

In general, most EP plasmas will have a flowing plasma component, but alignment of the probes such that $\boldsymbol{A} \perp \boldsymbol{U}_\infty$ (as in Fig. 7) ensures a negligible ram component $I_{\text{ion-sat}} \gg I_{\text{ion-ram}}$. One might be tempted to employ Eq. (20) as a method of measuring the plasma flow velocity, and indeed Mach probes (variations include Gundestrup, Janus, paddle, split Langmuir, and directional Langmuir) are Langmuir probes designed for this purpose [54–56]. The main difficulty in performing direct velocimetry with conventional Langmuir probes is that the ion density must be first computed in a manner that excludes any ram ion current.

Mach probes (see Fig. 8) bypass this limitation by using the ratio of ion currents from upstream and (flow blocked) downstream probes

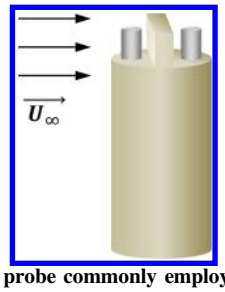

Fig. 8 Typical Mach probe commonly employed in tokamak plasma edge flow regions, uncommon and often inaccurate in EP plasmas.

(which cancels out the ion density term), but the theories are inexact [57,58] and limited, requiring numerical simulations to determine empirical fitting terms:

$$\frac{I_{\text{ion-upstream}}}{I_{\text{ion-downstream}}} = \exp\left(K\frac{U_\infty}{U_{\text{Bohm}}}\right), \qquad \frac{U_\infty}{U_{\text{Bohm}}} \lesssim 3$$
$$K = \begin{cases} 1.50\ T_i/T_e = 0.1 & \text{(unmagnetized)} \\ 1.35\ T_i/T_e = 1 & \text{(unmagnetized)} \\ 1/(0.43\sqrt{1+T_i/T_e}) & \text{(magnetized, } \boldsymbol{B}\|\boldsymbol{A}) \end{cases} \tag{21}$$

Simulations by Hutchinson are perhaps the most thorough, with relevant results summarized in Eq. (21). The complex wake (illustrated by Fig. 9) formed by a supersonic ion flow distorts the density and potential [59,60] significantly for downstream Mach probes, which is partly responsible for both the imprecise theory and limitation to only slightly supersonic flow ($U_\infty/U_{\text{Bohm}} \lesssim 3$).

Mach probes are often used (and have been validated extensively by Doppler-shift laser techniques and other methods) in subsonic, fully thermal, and fully magnetized flows. In the EP community, Mach probes are occasionally employed [61–64] but remain uncommon, perhaps limited by deteriorated accuracy for highly supersonic flows.

I. Magnetic Field Effects

Many EP thrusters use magnetic fields to keep the plasma away from thruster surfaces, to maintain accelerating electric fields, or to

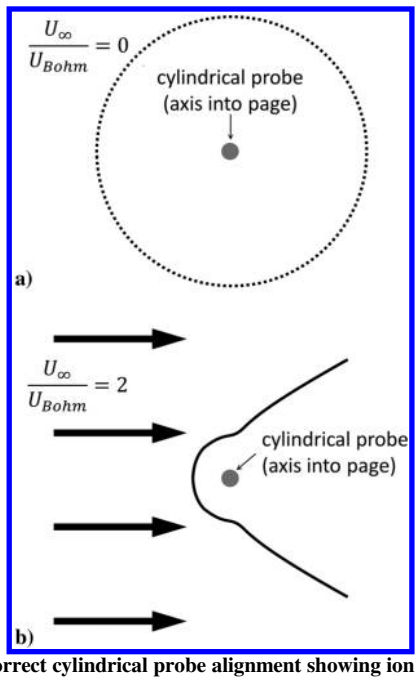

Fig. 9 Incorrect cylindrical probe alignment showing ion sheath wake zone: a) zero ion drifting speed, and b) ion drift speed twice the Bohm speed (data from simulations [59] for $(r_p/\lambda_D) = 1$ and bias $V_B = V_p - 25\cdot T_e$).

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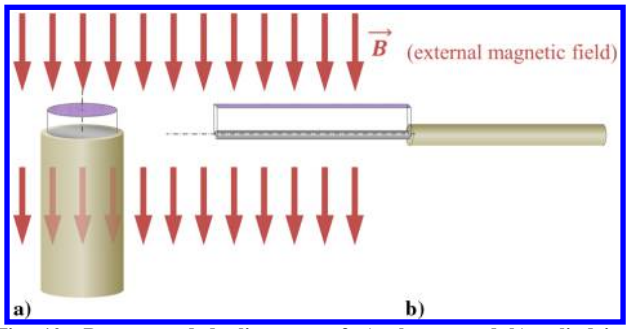

Fig. 10 Recommended alignment of a) planar, and b) cylindrical Langmuir probes in the presence of external magnetic fields showing effective electron current collection areas (shaded).

inductively couple to and accelerate a plasma. The effect of magnetic fields on the I–V characteristic is negligible in the limit of the probe radius being much smaller than the local Larmor radius, $r_{L,e} \gg r_p$. This is often achieved in many EP plasmas, and experiments for various magnetic field strengths (0–200 G) and field-to-probe angles show variation in plasma parameters within typical theoretical probe accuracies (see Sec. II.M) on the order of less than 50% [27]. However, by aligning the probe surfaces perpendicular to the external magnetic field, as in Fig. 10, the analysis of the electron flux (if magnetized) is simplified, and the distortion of the I–V curve should be reduced [11]. Indeed, to prevent depletion of electrons trapped on magnetic field lines (even for small probes with $r_{L,e} < r_p$), the probe orientation illustrated in Fig. 10 should be maintained whenever nonnegligible magnetic fields exist.

The effective area of collection for magnetized electrons is also reduced to the cross-sectional area of the probe that crosses the magnetic field lines [11], $A_{p,e-\text{eff.}} = \boldsymbol{A}_p \cdot \boldsymbol{B}/|B|$ (shaded offset regions in Fig. 10). Theoretical treatments of the magnetized electron flux variation with magnetic field strength exist [4,65] but are omitted here due to their inexactness and because the EP relevant plasma properties of density and electron temperature do not typically require such analysis. Magnetized current collection theories all rely on various terms to approximate (with limited accuracy or theoretical basis) anomalous cross-field electron transport processes. The collection of ion current is less affected by magnetic fields [3,4] (due to the heavier ions), with ion Larmor radii exceeding 1 m for typical EP beam ions in 200 G fields. Thus, the determination of ion density from ion saturation current is not likely to be affected by external magnetic fields especially at large negative probe biases (e.g., $V_b < V_p - 7.5T_e$ [41]) where magnetized electron current collection is negligible. Although ion current is usually immune and the electron temperature can still be determined from the inverse slope method [3,4,65] of $\ln I_e$ versus V_B (using $V_B \ll V_p$) in an electron-magnetized plasma, T_e can be overestimated by up to 30% [66], which can introduce some additional error into the ion density formulations [Eqs. (4, 6–8)] that depend of the square root of T_e or on powers of the electron Debye length. For more accurate magnetized plasma electron temperature determination, one could take the first moment [as in Eq. (13)] of the measured EEDF (using a magnetized EEDF formulation $\propto dI_e/dV_B$, as shown in [41]). The magnetization of the electrons can also obscure the I–V characteristic knee or transition from electron retarding to electron saturation [4,5], thus adding significant uncertainty in determining the plasma potential via conventional methods presented in Sec. II.F.

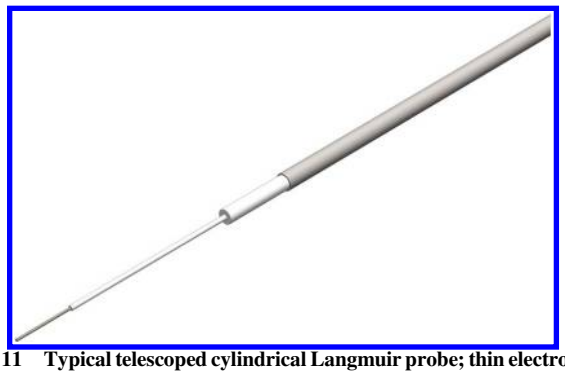

Fig. 11 Typical telescoped cylindrical Langmuir probe; thin electrodes, $L/r_p \gtrsim 50$, are necessary to avoid adverse end effects for cylindrical probes [19].

J. Langmuir Probe Construction

Langmuir probes are relatively easy to construct (see Fig. 11) because they simply require an exposed electrode held by an insulating support. Depending on the region, EP plasmas can lead to significant or negligible probe heating. Internal Hall thruster measurements can melt typical Langmuir probes in as little as 150 ms [67], thus requiring a high-speed positioning system (e.g., pneumatic piston or linear actuator). Probe ablation and secondary electron emission are two other undesirable effects that may occur in EP plasmas (due to the energetic ions and electrons). Refractory metals and ceramics are suitable probe construction materials that can withstand high temperatures, minimally erode, and produce few secondary electrons. In particular, tungsten, tantalum, or graphite electrodes and alumina (Al_2O_3) or boron-nitride (BN) insulating ceramics are commonly used in Langmuir probe construction with EP devices. Glass and plastic supports as well as soldered electrical connections to the electrode should be avoided because most glasses sputter easily upon energetic ion bombardment, whereas plastics and solder outgas or melt at relatively low temperatures. Many common materials outgas under vacuum conditions, and both NASA[‡] and the European Space Agency[§] maintain databases of the outgassing properties of various materials. Most common cable insulation is constructed of polyvinyl chloride, which is unsuitable for vacuum use. Teflon, silicon, and polyamide (e.g., Kapton) insulators are acceptable for vacuum use (where operating temperatures are within the material rating). Attention to cleanliness is also important during probe assembly, using powderless nitrile gloves when handling ceramics. Most ceramics are porous and will readily absorb oils and moisture from one's hands when handled without gloves. Although many of these contaminants should bake off in the vacuum environment or get ablated off under probe operation, the outgassing process could be lengthy and accompanied with the local generation of outgassing product gas that could skew probe measurements. Tungsten electrodes should be examined to have clean and smooth surfaces (light sanding with fine emery cloth or other smooth abrasive may be required to remove any oxide or contaminant layers). Gaps between the electrode, ceramic tubes, and metal tubing assembly should be minimized and filled with tape (e.g., Kapton or fiberglass tape) or ceramic paste (e.g., alumina-based, boron-nitride-based, or other compatible paste). Plasma will easily enter any gap bigger than a Debye length, but accelerated ions can enter any sized gap along their trajectory path. Sample photos of cylindrical and planar probes used in EP research are shown in Fig. 12 (the planar probe here is unguarded; a guarded probe as in Fig. 5 is recommended because it will have less sheath expansion and insulator wall effects). Just before first postassembly operation, a Langmuir probe electrode should be biased (while under vacuum with a plasma discharge present) to a very large bias (e.g., −100 V) until the probe begins to glow (light red or orange, not white) because this will ablate off most contaminants (oils, sputtered glasses, etc.). Xenon and other noble gas propellants will not directly react with Langmuir probe electrodes but can lead to slow erosion; hence, pre- and postoperation probe dimension measurements are recommended to assess any probe degradation during testing. Noble gases can, however, sputter contaminants onto the Langmuir probe collection electrode, and in such environments (such as an EP device using quartz tube discharge), frequent probe cleaning during testing may be required (by using a large bias). A contaminated probe surface can increase the resistance of the electrode and lead to significant measurement error including falsely

[‡]Data available online at http://outgassing.nasa.gov [retrieved 07 December 2013].

[§]Data available online at http://esmat.esa.int [retrieved 07 December 2013].

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Fig. 12 Photographs of cylindrical (upper) and planar (lower) Langmuir probes used in EP plasmas.

high electron temperatures. Electrical connections to the collection tungsten electrode should be crimped to ensure reliable conduction through large temperature swings.

K. Radio-Frequency-Compensated Langmuir Probes

Determining the time-averaged plasma properties in a highly fluctuating plasma is not as simple as measuring the time-averaged I–V characteristic. In capacitive radio-frequency (RF) discharges, the plasma (and floating) potential can swing ± 100 V or more [68] during a single RF cycle, and the time-averaged I–V curve from a noncompensated Langmuir probe would exhibit irrevocable distortion. The use of RF-compensated Langmuir probes [11,68] (Fig. 13) is one solution to this issue. These probes rely on a series of self-resonating inductors (notch filters tuned to the primary frequency f_{RF} and the next few harmonics $2f_{\mathrm{RF}}$ and $3f_{\mathrm{RF}}$) placed in close proximity to the Langmuir probe electrode. Another (much larger) exposed reference electrode is used to capacitively couple local plasma fluctuations (with capacitor $C_{\mathrm{rc}} \approx 1$–$3$ nF typical) with the notch filters to lower the high-frequency loading of the probe electrode. Small surface mount inductors $L_{1\text{–}3}$ are used, and the capacitances $C_{1\text{–}3}$ are the stray interwinding capacitance of those inductors (or tiny variable capacitors to allow manually tuning).

RF-compensated Langmuir probes have shown moderate success [69] in measuring the time-averaged plasma properties in many RF plasmas (such capacitive 13.56 MHz discharges), but the realization of effective high-frequency filter components can prove challenging (owing to size constraints and finite stray capacitance).

L. High-Speed Dual Langmuir Probes

Large-amplitude plasma oscillations are known to distort the time-averaged I–V characteristic and can lead to significant errors in the computation of the plasma properties from such I–V curves. Although RF-compensated probes can alleviate this issue in fixed-frequency RF discharges (e.g., 13.56 MHz), the filtering techniques are too narrow-band to subdue most natural stochastic (turbulent) plasma fluctuations (e.g., breathing mode, spoke modes, etc., encountered in Hall thruster discharges). One solution is to sweep the probe bias rapidly enough that the plasma properties are stable throughout the duration of the sweep (as seen in rapidly sampled but smooth I–V traces in Fig. 14). A high-speed dual Langmuir probe (HDLP) [70] is designed to operate with bias sweep rates from 100 kHz to 1 MHz using a single Langmuir probe (and the same theory of analysis) in conjunction with a null compensation probe in close proximity.

Fabrication of an HDLP is similar to a single Langmuir probe but with the use of double-bore alumina (or other ceramic) tubing to

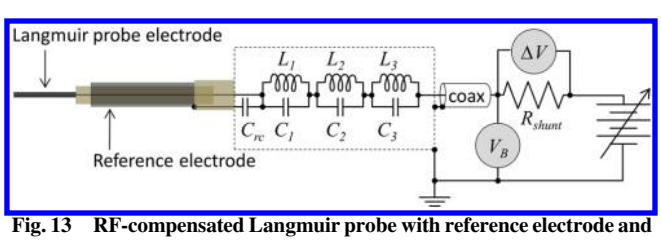

Fig. 13 RF-compensated Langmuir probe with reference electrode and three-stage notch filters to enable time-averaged measurements for a fixed RF plasma generation frequency $f_{\mathrm{rf}} \approx 1/(2\pi\sqrt{L_1 C_1})$.

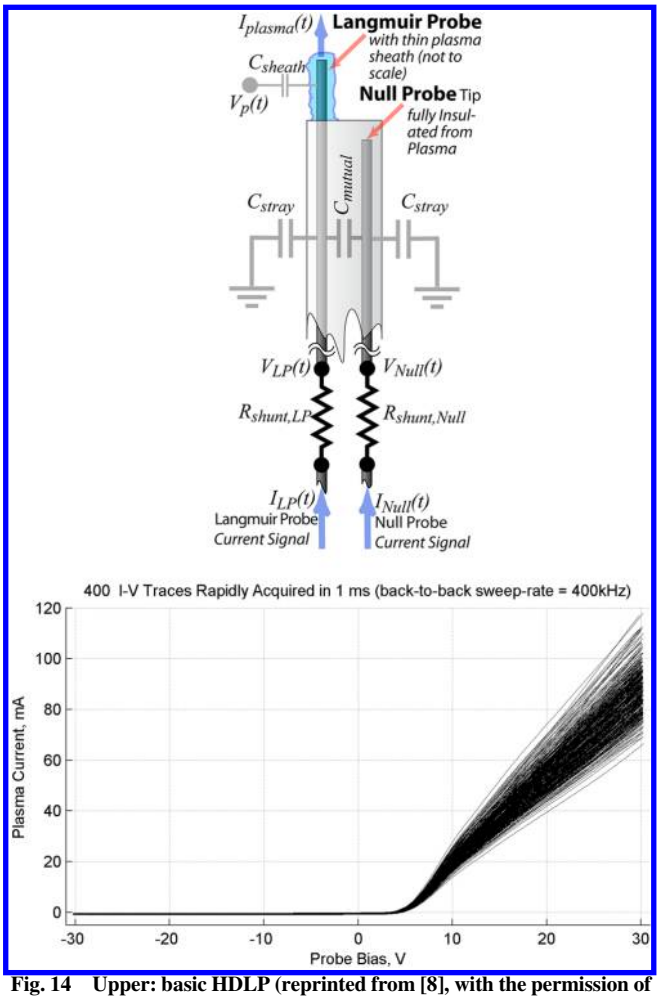

Fig. 14 Upper: basic HDLP (reprinted from [8], with the permission of AIP Publishing). Lower plot (from [71]) shows 400 I–V characteristics collected by an HDLP downstream from a Hall thruster.

facilitate the close spacing of the active Langmuir probe electrode and the fully insulated null probe. However, HDLP measurement electronics are more complex because the current and voltage sensing bandwidth required is on the order of dc to 50 MHz (to collect enough I and V pairs during each brief 1–10 μs sweep duration). The dc response is needed to accurately capture the flat ion saturation portion of the I–V characteristic, whereas the higher-frequency ($\approx$50 MHz) requirement arises from the sharp (exponential) current transition in the electron retarding region. Voltage is relatively easy to measure with this bandwidth, but current (especially the small currents riding on the high probe bias voltages) is more challenging and requires a custom setup. The shunt resistors in Fig. 14 can be used directly (with voltage dividers, buffers, and differential amplifiers) to measure the current as a voltage drop (as in Fig. 1), but the value of the resistor should be chosen to match the transmission line impedance of the cabling (e.g., 50 Ω noninductive shunts for standard RG-58, RG-316, or RG-188 coaxial cables) to each the active and null probes. High-frequency differential current measurement, isolated from the high probe bias voltages, is possible with a differentially wound Pearson coil [71] (multiple windings of the Langmuir probe signal line and counterwindings of the null probe signal line). Because Pearson coils cannot measure dc currents, the voltage drop across the shunts should be recorded with a digital multimeter (DMM) or other lower-bandwidth differential voltage measurement. As is emphasized in Sec. V, the accuracy of the current measurement must be sufficient to capture all portions of the I–V characteristic; this constraint precludes the use of oscilloscopes (which predominantly employ 8-bit flash-based analog-to-digital convertors) and typically requires the use of a dedicated high-speed 16-bit (or higher) data acquisition system.

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M. Limitations of Single Langmuir Probes

Langmuir probes always perturb the plasma environment because they must be physically placed where a measurement is desired. Keeping probes small limits the physical space they must occupy but even an infinitesimally small probe can create a sheath up to 10 or more Debye lengths in size about itself. The other main limitations of Langmuir probes are the assumptions required by the theories applied to interpret the probe I–V characteristics into meaningful plasma properties.

Experimental efforts studying the accuracy of Langmuir probes by means of comparing against other validated measurement techniques (i.e., microwave cavity and interferometry, hairpin resonant probes, radar scattering, spectroscopic, and other methods) have shown agreement within 3–30% for n_e [72–75], within 10–50% for n_i [69,72,76], within 2–11% for T_e [77–79], and within 12–20% for V_p [80]. These studies suggest a variety of possible mechanisms that can lead to errors in Langmuir probe measurements, including (but not limited to) secondary electron emission or reflection, photoelectric emission, contaminated probe or support surfaces, probe perturbation of plasma, finite probe size end effects, and plasma fluctuations. An HDLP eliminates negative effects of most plasma fluctuations by sampling the plasma much faster the time scale of the bulk plasma fluctuations, but the maximum theoretical HDLP sweep rate is on the order of 1 MHz for EP plasmas [8] with 400 kHz systems in use today [71].

III. Double Langmuir Probe

A. Theory

There exist several variations of Langmuir probe implementations, which may prove advantageous for certain experimental situations. One such implementation is the double probe (Fig. 15), which is essentially a pair of Langmuir probe electrodes connected together as shown in Fig. 16. Note that, for the case of the double probe, the bias voltage V_B is applied between two probe electrodes, unlike the single Langmuir probe, where V_B is applied between the probe and a stable ground reference electrode (e.g., vacuum chamber walls). This distinction is one of the key advantages of the double-probe configuration; its application does not require the presence of a stable ground reference nor does it require knowledge of the local plasma potential. In addition, because the probe as a whole floats, there is no net current drawn from the surrounding plasma. Continuity limits the total current through the probe to the smaller of the currents that can be collected by either individual electrode (i.e., the ion saturation current). Because of this, the current through the probe is small compared to that of a single Langmuir probe (LP) operating in the electron attracting regime and heating of the probe electrodes is reduced, as is the likelihood of disturbance to the EP device under test.

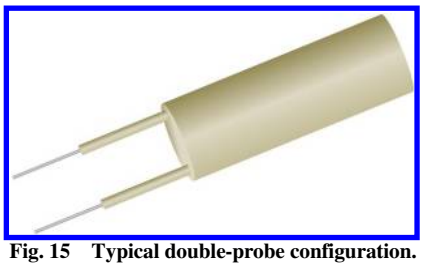

Fig. 15 Typical double-probe configuration.

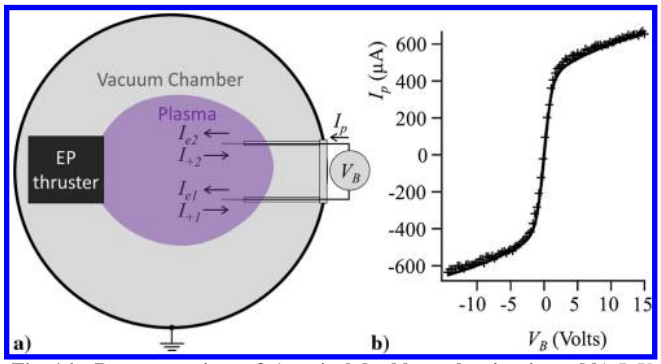

Fig. 16 Representations of a) typical double-probe circuit, and b) I–V characteristic for symmetric double-probe pair in the plume of a Hall thruster (reprinted from [20], with the permission of AIP Publishing).

The double-probe characteristic arises by applying Kirchhoff's laws, which lead to Eqs. (22) and (23), and by noting the electron current collected by a biased electrode as given by Eq. (24). Manipulation of these relations leads to the somewhat cumbersome expression given as Eq. (25). A common implementation of double probes is referred to as the symmetric case ($R = 1$, equal electrode areas), which results in the compact form given by Eq. (26). With this fundamental probe characteristic, interpretation of double-probe data relies upon appropriate application of an ion current collection relation. In principle, any of the ion current collection models described previously (in Secs. II.B–II.D) can be applied to the double probe. For example, for large probes and very dense plasmas, it is reasonable to approximate the ion current as the value calculated from the thin-sheath relation given by Eq. (4). In this approximation, the ion current is taken to be independent of the electrode potential such that $I_{+1} = I_{+2}$ and Eq. (26) reduces to a simple hyperbolic tangent expression. Ion density and electron temperature can then be estimated by choosing the values of T_e and I_{+1} (or n_i) that give the best fit of Eq. (26) to experimental I–V data:

$$I_P = I_{+1} - I_{e1} = I_{e2} - I_{+2} \tag{22}$$

$$V_B = V_2 - V_1 \tag{23}$$

$$I_{\mathrm{en}}(V_n) = I_{\mathrm{en,sat}} \exp\left(\frac{V_n}{T_e}\right), \text{ with } I_{\mathrm{en,sat}} = e n_o A_n \sqrt{\frac{eT_e}{2\pi m_e}} \tag{24}$$

$$I_P = \frac{I_{+1}[R\exp(V_B/T_e) - 1]}{[R\exp(V_B/T_e) + 1]} + \frac{I_{+1} - I_{+2}}{[R\exp(V_B/T_e) + 1]} \tag{25}$$

$$I_P = I_{+1}\tanh\left(\frac{V_B}{2T_e}\right) + \frac{I_{+1} - I_{+2}}{\exp(V_B/T_e) + 1}, \quad \text{for } R = A_2/A_1 = 1 \tag{26}$$

Although the simple approach described previously can be used successfully when the plasma sheath is small with respect to the double-probe electrode diameters (namely, when $r_p/\lambda_D \gtrsim 50$), most laboratory experiments will benefit from a more versatile approach that does not rely on such a limiting assumption. Such versatility can be achieved by applying a more accurate description of the ion currents in Eq. (26) [or Eq. (25) for the asymmetric double probe]. This can be readily accomplished by using fits to Laframboise's results of the form given by Eq. (27), where the values of a and b are given by Eq. (7) [20]. Combining Eq. (27) with Eq. (26) yields Eq. (28), which gives an accurate description of probe current for any ratio of sheath thickness to electrode size. To use this relation, one must first determine the value of V_1 in terms of directly measurable quantities such as V_B. This is readily achieved by noting that the probe as a whole floats (i.e., $I_{+1} + I_{+2} - I_{e1} - I_{e2} = 0$). Using Eq. (27) for the ion currents and Eq. (24) for the electron currents, V_1 can be determined implicitly as a function of V_B for given values of electron temperature, a, and b:

$$I_{+n} = e^{3/2} n_o A_n \left(\frac{T_e}{2\pi m_i}\right)^{1/2} a\left(\frac{-V_n}{T_e}\right)^b \tag{27}$$

$$I_P = e^{3/2} n_o A_n \left(\frac{T_e}{2\pi m_i}\right)^{1/2} \left[a\left(\frac{-V_1}{T_e}\right)^b \tanh\left(\frac{V_B}{2T_e}\right) + \frac{a(-V_1/T_e)^b - a(-(V_1+V_B)/T_e)^b}{\exp(V_B/T_e)+1} \right] \tag{28}$$

Raw double-probe I–V characteristics (as Fig. 16b portrays) can be used to determine electron density and electron temperature using the preceding relations and an iterative approach. First, rough estimates of n_o (note that $n_o \approx n_e \approx n_i$) and T_e are determined using the thin-sheath approximation, and these values are in turn used to estimate the a and b parameters. Standard numerical root finding techniques are then used to determine V_1 as a function of V_B. These values are then inserted into Eq. (28), and a curve-fitting routine, such as the common Levenberg–Marquardt method, is used to determine updated values of n_i and T_e that provide the best fit to the experimental data. The procedure then continues iteratively until it converges on a self-consistent set of plasma parameters. Examples showing the application of this technique can be found in [20].

B. Construction

Double probes can be constructed using the method described in Sec. II.J, with the obvious exception that two electrodes are required. The double probe can be constructed by simply wiring together two single probes or by the slightly more elegant approach of using double-bore alumina tubes as the insulator material to hold both electrodes. In either case, care must be taken to space the electrodes appropriately according to two competing requirements. First, the electrodes must be close enough together such that both are exposed to identical plasma conditions. Second, the electrodes must be sufficiently far apart to prevent overlap of the plasma sheaths.

The most common geometry for double probes, and the one that is recommended for use in flowing plasmas, is the symmetric probe using two identical cylindrical electrodes as shown in Fig. 15. The cylindrical electrodes should be oriented parallel to the predominant plasma flow direction to minimize the effect of directed flux on inferred plasma density (see Fig. 7).

C. Limitations

In addition to the previously described limitations that are inherent to all Langmuir probe configurations, a key disadvantage of the double probe is its physical size. Because it relies upon two electrodes, it is inherently larger than a single LP, and the plasma measurement location includes the space occupied by both electrodes. It is therefore important that the double probe be sized small compared to the spatial scale length of the plasma plume such that the plasma parameters can be considered identical at both electrode locations. For the symmetric double probe, violation of this requirement will manifest as asymmetries in the double-probe characteristic of Eq. (26).

The other key limitation of the double probe is that its interpretation provides no information about the electron energy distribution function. Rather, it assumes the presence of a single, Maxwellian electron population. Because both electrodes operate well below the plasma potential, in the vicinity of the floating potential, only the most energetic electrons in the distribution are collected. In the case of a plasma containing two distinct electron populations, the electron temperature reported by the double probe will reflect the tail population (which could be "hotter" or, if depleted, "cooler") of the two distributions, which can lead to an over- or under-estimate of the bulk electron temperature [81]. An example of where this issue may be seen would be near the cathode of an EP device where a population of plume electrons may coexist with a distribution of cathode electrons. For this reason, double probes are best suited to situations where the assumption of a single electron population is justified. A somewhat more accurate electron temperature measurement may be obtained from an asymmetric double Langmuir probe (where the more negative probe is much larger than the other probe) because a broader portion of the single probe I–V characteristic is captured [4,5,82].

IV. Triple Langmuir Probe

A. Theory

A configuration of Langmuir probe known as the symmetric triple probe consists of three equal-sized electrodes configured as shown in Fig. 17 [83]. In Fig. 17, electrode 2 is shown to be electrically floating (i.e., $I_2 = 0$). By applying Kirchhoff's laws and the previously stated expression for electron current [Eq. (24)], the fundamental triple-probe relation can be expressed as [83]

$$\frac{I_1 + I_2}{I_1 + I_3} = \frac{I_{+2} - I_{+1} + I_{e1,\mathrm{sat}} \exp(V_1/T_e)[1 - \exp(V_{d2}/T_e)]}{I_{+3} - I_{+1} + I_{e1,\mathrm{sat}} \exp(V_1/T_e)[1 - \exp(V_B/T_e)]} = \frac{1}{2} \tag{29}$$

In principle, any of the previously described ion current models can be applied to Eq. (29) and the triple probe. In practice, however, the primary advantage of the triple-probe geometry is its simplicity of operation, which allows measurement of plasma density and electron temperature using a fixed voltage differential rather than via interpretation of a characteristic function resulting from a voltage sweep. To facilitate this simplicity, the assumption of constant ion current ($I_{+1} = I_{+2} = I_{+3}$) is usually applied, which allows the triple-probe equation to be simplified to

$$\frac{[1 - \exp(V_{d2}/T_e)]}{[1 - \exp(V_B/T_e)]} = \frac{1}{2} \tag{30}$$

Although the assumption of equal ion currents to each electrode is not rigorously accurate, the error introduced by this approximation can be minimized by limiting the constant voltage applied between electrodes 1 and 3, V_B, to a value on the order of the electron temperature. Such a selection of V_B ensures that all three electrodes operate in reasonable proximity to the floating potential such that they collect ion currents that vary by only a small amount, which can be quantified by proper application of Eq. (7). With these precautions in place, the triple probe immediately provides the electron temperature [via Eq. (30)] using just the applied voltage V_B and the measured potential V_{d2} between the floating electrode and electrode 1. The plasma number density, equal to the ion density in the bulk plasma outside the sheath, can then be calculated from the measured probe current using the Bohm condition according to the following relation, where the probe surface area A_1 should be replaced by the sheath surface area in situations where the Debye length is not negligible relative to the probe size:

$$n_i = \left(\frac{m_i}{T_e}\right) \frac{I_p \exp(1/2)}{A_1 e^{3/2}[\exp(V_{d2}/T_e) - 1]} \tag{31}$$

Because the triple probe does not require a voltage sweep, but rather the simultaneous measurement of several voltages and a single

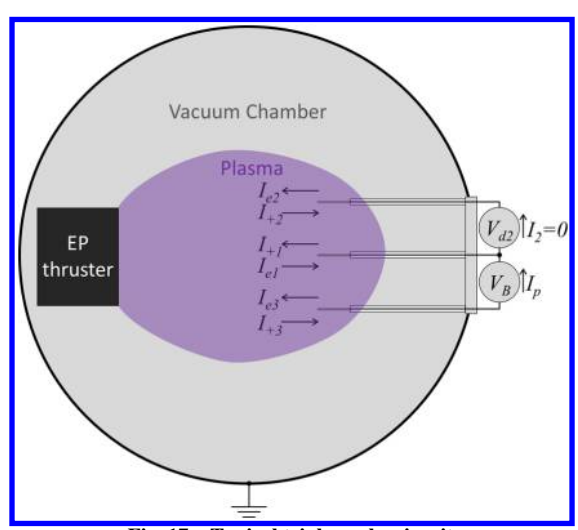

Fig. 17 Typical triple probe circuit.

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current, this technique is particularly well suited to use in time-varying plasmas. In such applications, care must be taken to ensure that the temporal response of the measurement circuit is sufficiently fast to track the plasma time scales of interest. In addition to time-varying plasmas, triple probes are also readily applied to flowing plasmas such as EP plumes [84–87].

B. Construction

Triple-probe construction can be performed in a manner identical to that of the double probe, with the addition of an additional electrode. Like the double probe, care must be taken to ensure that the electrodes are sufficiently separated so as to avoid sheath overlap but adequately close to each other such that they can be approximated as sampling identical plasma conditions.

C. Limitations

The feature that makes the triple probe attractive and simple to employ also leads to its most significant limitation. Because the device does not employ a voltage sweep, the amount of information collected about the plasma is limited. A complete triple-probe measurement is analogous to obtaining three I–V pairs on the Langmuir probe characteristic: the floating potential, one point below the floating potential at a finite current, and one above the floating potential at an equal current magnitude. These three points are completely sufficient for determination of electron temperature and density in a well-behaved, Maxwellian plasma, but under nonequilibrium plasma conditions, the triple probe offers no information about the electron energy distribution function, nor does it offer clues that the population is non-Maxwellian. In addition, because this technique does not involve fitting to a characteristic equation, there is no inherent noise rejection, and the triple probe can be susceptible to measurement noise. This limitation is usually easily overcome by averaging multiple measurements. Because of these limitations, the triple probe is best suited to situations where the plasma can reasonably be assumed to be Maxwellian.

V. Data Collection, Calibration, and Error Analysis

Langmuir probe determined plasma property uncertainties (prefixed with "∂") associated with the accuracy of the voltage (∂V_B), current (∂I_p), and probe size ($\partial r_p, \partial L$) measurements are easily quantified through standard error propagation methods and are presented in Eq. (32) [70]:

$$\begin{aligned}
\frac{\partial A_p}{A_p} &\approx \frac{2\pi}{A_p}\sqrt{(L+r_p)^2\partial r_p^2 + r_p^2\partial L^2} \\
\frac{\partial T_e}{T_e} &\approx \sqrt{2\left(\frac{\partial V_B}{\Delta V_B}\right)^2 + 2\left(\frac{\partial I_{e,\mathrm{sat}}}{I_{e,\mathrm{sat}}}\cdot\frac{T_e}{\Delta V_B}\right)^2} \\
\frac{\partial n_e}{n_e} &\approx \sqrt{\left(\frac{\partial I_{e,\mathrm{sat}}}{I_{e,\mathrm{sat}}}\right)^2 + \left(\frac{\partial A_p}{A_p}\right)^2 + \left(\frac{\partial T_e}{2T_e}\right)^2} \\
\frac{\partial n_i}{n_i} &\approx \sqrt{\left(\frac{\partial I_{i,\mathrm{sat}}}{I_{i,\mathrm{sat}}}\right)^2 + \left(\frac{\partial A_s}{A_s}\right)^2 + \left(\frac{\partial T_e}{2T_e}\right)^2} \\
\frac{\partial V_p}{V_p} &\approx \frac{1}{V_p}\sqrt{(\partial V_B)^2 + (\Delta V_B)^2} \\
\frac{\partial V_f}{V_f} &\approx \frac{1}{V_f}\sqrt{(\partial V_B)^2 + \left(T_e\frac{\partial I_{i,\mathrm{sat}}}{I_{i,\mathrm{sat}}}\right)^2}
\end{aligned} \tag{32}$$

This uncertainty formulation includes the direct measurement errors as well the discretized nature of the collected data. The finite spacing between bias points is denoted as ΔV_B, and two current measurement uncertainties are included, $(\partial I_{e,\mathrm{sat}}/I_{e,\mathrm{sat}})$ and $(\partial I_{i,\mathrm{sat}}/I_{i,\mathrm{sat}})$, to account for differing instrument accuracies at these current levels. For a typical Langmuir probe (0.5 mm radius and 10 mm length) in a Hall thruster plume with typical instrument accuracies of $\pm 1\%$ for voltage, probe length, and electron current, $\pm 5\%$ for probe radius, and $\pm 10\%$ for ion current, the approximate plasma uncertainties are $(\partial T_e/T_e) = \pm 13\%$, $(\partial n_e/n_e) = \pm 8\%$, $(\partial n_i/n_i) = \pm 54\%$, $(\partial V_p/V_p) = \pm 2\%$, and $(\partial V_f/V_f) = \pm 20\%$. Included in this calculation is a sheath area uncertainty set to 10 times the probe area uncertainty, thus giving rise to the low ion density accuracy. This collection of error estimates neglects the uncertainty or validity of the theoretical analysis presented in Sec. II.M, which can be larger, particularly for ion current collection theories [88]. Poor measurement setup can lead to significant errors especially in the measurement of current. The use of a single measurement range setting for the current measurement instrument is often problematic because the ion and electron saturation currents differ substantially ($I_{e,\mathrm{sat}} \approx -323 \cdot I_{i,\mathrm{sat}}$ for xenon). For example, a typical oscilloscope (with 8 bits of measurement resolution) set to capture electron saturation will have a bit resolution of $\pm I_{e,\mathrm{sat}}/2^8$, which corresponds to a minimum current uncertainty of $\pm 126\%$ of $I_{i,\mathrm{sat}}$. Also problematic is the use of two single-ended measurements to determine the voltage drop ΔV across the shunt resistor because this voltage is typically much smaller than the bias voltage (for recommended shunts resistances $\leq 100\ \Omega$), thereby requiring a large voltage measurement range with uncertainty possibly comparable to ΔV. A dedicated differential voltage measurement of ΔV is recommended using either a DMM (with + and − inputs across shunt) or an instrumentation amplifier with differential inputs. Another common issue is leakage current that has not been accounted for that flows into the measurement instrument due to finite input impedance as illustrated in Fig. 18a. This leakage current is approximately equal to the bias voltage divided by the voltage measurement input impedance (noting that Z_{input} is not always a constant). Common input impedances are 1–10 MΩ, which corresponds to −50 to −5 μA of current leakage at a −50 V bias, current values that could easily exceed the local ion saturation current. Three solutions are 1) remove this leakage current in postprocessing as shown in Fig. 18a, 2) use a voltage measurement instrument with a "high-Z" mode where $Z_{\mathrm{input}} \gg$ GΩ, or 3) use a source measurement unit (SMU) as shown in Fig. 18b. The SMU

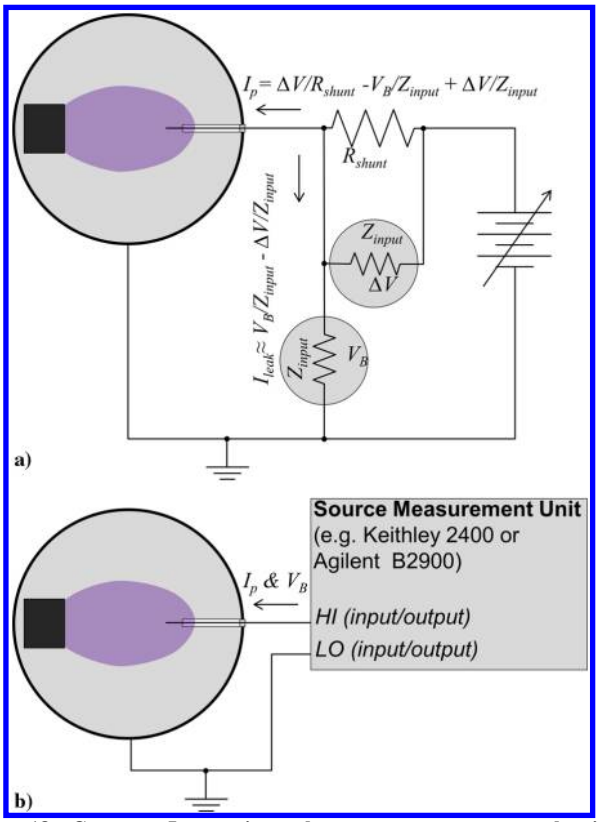

Fig. 18 Common Langmuir probe measurement setups showing a) leakage current accounting from measurement input impedance Z_{input}, and b) use of an SMU to source V_B and measure the resulting I_p.

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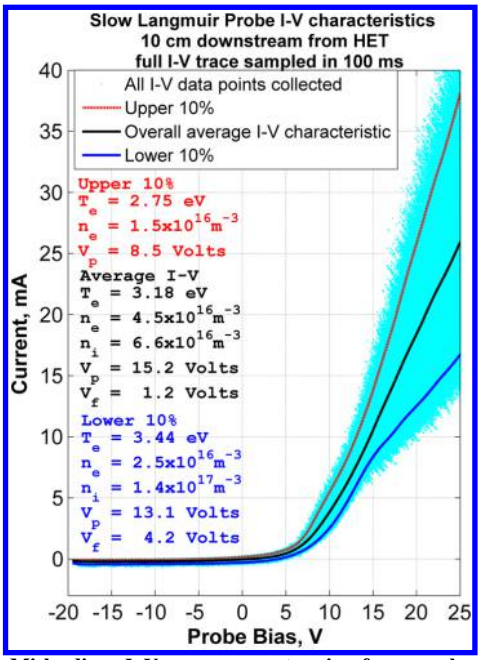

Fig. 19 Misleading *I–V* measurement noise from a slowly swept Langmuir probe downstream from a Hall thruster (from [89]).

approach is recommended because these devices are readily interfaced (via general purpose interface bus (GPIB), USB, or Ethernet) using a computer to control the bias voltage scan range and record the measured voltage and current with excellent precision and accuracy. Many SMUs and DMMs can be run in autoranging mode that allows for vastly improved accuracy in the measurement of both small $I_{i,\mathrm{sat}}$ and large $I_{e,\mathrm{sat}}$ currents. SMUs and DMMs often also have a measurement integration time setting that is useful in reducing the measurement noise of the nonquiescent plasmas typical of EP discharges.

Sampling an *I–V* characteristic without any averaging or filtering reveals the extent of plasma fluctuations that appear as "noise" from the scatter of *I–V* points shown in Fig. 19 [89]. This *I–V* trace was acquired in 100 ms and includes several thousand breathing mode fluctuation cycles. Rapidly sampling an *I–V* trace with a HDLP (in 10 μs or faster) removes the apparent noise with qualitatively smooth *I–V* curves as seen in Fig. 14.

Another important consideration to maintain acceptable measurement accuracy is to keep all instruments fully calibrated. SMUs and DMMs usually have six or more measurement ranges, and each one requires periodic factory (or user) calibration to maintain the manufacturer specified accuracies. Failure to sustain calibration can lead to peculiar jumps in the apparent voltage or current as the instrument automatically switches measurement ranges. Shunt resistors should be calibrated as well (often a four-point measurement with a calibrated DMM will suffice), and the power ratings of these shunts must be observed to maintain low thermal drifting errors (most resistors drift in excess of 100 ppm/°C).

VI. Conclusions

A brief guide has been presented that outlines the recommended methods for designing, operating, and analyzing Langmuir probes for use in electric propulsion (EP) plasma environments. The basic equations of current collection for single, double, triple, and other (Mach, high-speed dual Langmuir probe, etc.) Langmuir probes have been included with their various operational regimes and assumptions discussed. The relative simplicity of Langmuir probes, coupled with the wealth of information of the state of the plasma that they provide, make them an indispensable plasma diagnostic for the study of EP devices. The recommended practice herein provided should help an experimentalist mitigate many of the more common mistakes of Langmuir probe operation in EP devices and ensure accuracies comparable to other plasma diagnostic techniques.

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